

IS-003 Sentinel Event Bus

Implementation Specification IS-003 - Sentinel Event Bus

Status: Approved for Design

Priority: Critical

Package ID: PKG-003

Depends on: IS-001, IS-002

Referenced Blueprint Volumes: 33, 34, 56

Vision

Everything that happens inside Project Sentinel becomes an event. Events are the nervous system of the Cyber Security Operating Environment.

Event Principles

Events are immutable, timestamped, traceable, correlated, serializable, versioned and observable.

Event Structure

Event ID, Event Type, Object ID, Correlation ID, Timestamp, Source Engine, Actor, Payload, Version, Priority, Classification, Signature (future), Metadata.

Lifecycle

Created -> Validated -> Published -> Received -> Processed -> Archived

Requirements

REQ-EVENT-001: Every event shall have a globally unique Event ID.

REQ-EVENT-002: Every event shall be immutable after publication.

REQ-EVENT-003: Every event shall include a timestamp and correlation ID where applicable.

REQ-EVENT-004: The Event Bus shall support multiple subscribers.

REQ-EVENT-005: Subscribers shall filter events without modifying them.

REQ-EVENT-006: Event delivery failures shall be observable and recoverable.

REQ-EVENT-007: Security classification shall be enforced during event delivery.